

EdgeHealth Sentinel — Adaptive, Uncertainty-Aware and Equity-Driven Edge AI for Health-System Intelligence

1. Challenge Tackled

I built EdgeHealth Sentinel to explore a practical problem in resource-constrained healthcare: how can a health system make better decisions when it does not have enough evidence about the communities it serves? Limited testing, health workers, equipment, connectivity, and transportation can create health-system blind spots. The absence of data should not automatically be interpreted as the absence of need. The prototype focuses on malaria as a concrete use case while also looking at the wider health system around the diagnosis. It is designed for health workers and decision-makers who need to understand both what the AI sees and what the system does not know.

2. Tools / ML Models Used

- Python 3.13 — data processing and ML pipeline
- Streamlit — interactive application/dashboard
- scikit-learn — machine learning
- HOG + Logistic Regression — lightweight malaria blood-smear image classification
- scikit-image / Pillow — image processing
- Joblib — model persistence
- Synthetic community data — health-system, resource, geographic, and equity-related simulation
- What-If simulation — tests how resource/testing changes affect selected system indicators

Architecture: Quality Gate → Model → Uncertainty → Safety Governor → Human Decision

3. What Worked Well

The strongest part is that the AI prediction is not treated as a final answer. The system assesses image quality and uncertainty and recommends additional evidence or human review when the result is not sufficiently reliable. At community level, the prototype makes evidence debt and geographic blind spots visible. The What-If simulator lets users explore possible resource changes instead of only showing a static prediction. The prototype is deployed as a working Streamlit app with a public GitHub repository.

4. What Was Challenging

The biggest challenge was connecting the ML component with a meaningful health-system decision layer without making unsupported clinical claims. I also had to recover and stabilize the Streamlit application after a code-file issue, then test locally and deploy it. Community and equity analysis uses synthetic data; these indicators are for demonstration and simulation, not validated population-level measurements.

5. How I Spent My Hackathon Time

- Initial: define the healthcare problem and core concept
- Build: malaria image pipeline and health-system intelligence
- Integration: diagnostic safety, resource analysis, equity-related indicators, and What-If simulation
- Debugging: test the app, fix integration issues, stabilize deployment
- Final: public repository, documentation, demo, technical video, and submission

Reflection

If I had more time, I would expand validation with larger and more representative datasets, improve model calibration and evaluation, and test with real healthcare stakeholders. **AI proposes. Evidence constrains. Simulation tests. Humans decide.**

Research prototype — not a clinically validated diagnostic system. Region X and community-level simulation outputs are synthetic.